Panic Selling When Disaster Strikes: Evidence in the Bond and Stock Markets

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Question

- How do stock and bond markets respond when firms are exposed to natural disasters?
- Are higher future returns driven by risk compensation or panic selling?

Why interesting?

- **Physical climate risk:** Extreme climate events cause large economic losses and disrupt firm operations.
- **Pricing remains unclear:** Disaster exposure raise firms' financing costs?
  - Prior studies mainly focus on ESG and carbon emissions.
- **Firm-level exposure is hard to measure precisely.**
  - Headquarters locations may not reflect actual operating exposure.
  - Production, sales, and supply chains are often geographically dispersed.
- **This study:** Use establishment-level data to measure disaster exposure and examine its pricing in stock and bond markets.

Contribution

- **Extending physical climate risk pricing**
  - Prior studies focus on carbon risk, climate regulatory risk, ESG, and the operational effects of natural disasters (Engle et al.,2020)
  - This study examines how firms' direct disaster exposure is priced in both stock and corporate bond markets.

- **Improving the measurement of disaster exposure**
  - Alok et al.(2020) identify firms' exposure using headquarters locations.
  - This study matches establishment-level locations from NETS with county-level disaster data from SHELDUS, providing a more precise measure of firms' direct exposure.

Design–Hypotheses

1. **H1: Future returns.** Disaster exposure is positively associated with one-month-ahead stock and bond returns.
2. **H2: Contemporaneous returns.** If investors overreact, disaster exposure is negatively associated with contemporaneous returns.
3. **H3: Insider purchases.** If investors overreact, insiders' share purchases increase following disaster exposure.
4. **H4: Investor attention.** The effect of disasters is more pronounced after 2006.
5. **H5: Environmental performance**. The effect is stronger for firms with low environmental performance, fewer sustainability-oriented investors, and more short-term investors.

Design–Natural Disaster

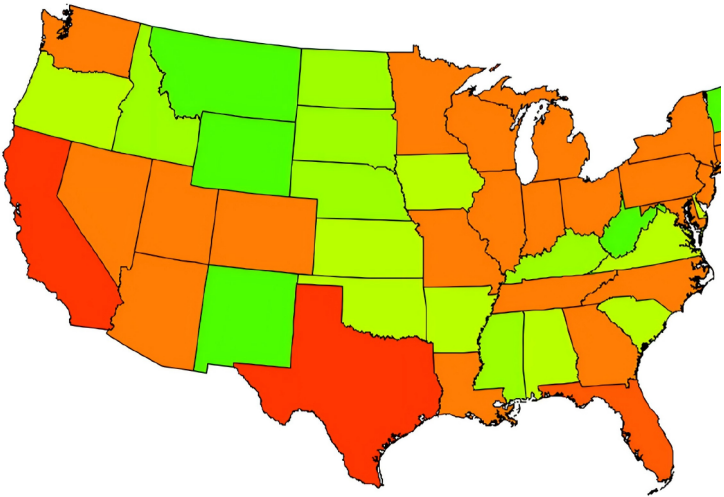
- **Disaster data:**
  - SHELDUS:retains disasters lasting less than 30 days and exceed $10 million
- **Establishment matching:**
  - locations of firms' plants, branches, and subsidiaries are obtained from NETS.
- **Main measure:**

$$Disaster_{i,t} = \begin{cases} 1, & \text{if at least one plant of firm } i \text{ is located in an affected county;} \\ 0, & \text{otherwise.} \end{cases}$$

- **Caontinuous measure:**

$$Ln(Damage_{i,t}) = \ln\left(1 + \sum_{c} w_{i,c} Damage_{c,t}\right)$$

  - where $w_{i,c}$ is the establishment-sales weight.
- **Sample cleaning:** Financial firms are excluded.



- Locations of Establishments Exposed to Natural Disasters Between 1990 and 2015

Design–Baseline Regression

- The main panel regression is:

$$ExcessReturn_{i,t+1} = \alpha + \gamma Disaster_{i,t} + \delta' X_{i,t} + \theta_i + \phi_y + \psi_m + \varepsilon_{i,t}.$$

- $ExcessReturn_{i,t+1}$ is the one-month-ahead stock or bond excess return.
- $X_{i,t}$ includes standard stock or bond return predictors.
- Security, year, and month fixed effects are included.
- $\gamma > 0$ is consistent with both higher risk compensation and price reversal.

Design–Overreaction

- **Fundamentals:** disasters should reduce establishment and firm sales.
- **Return dynamics:**

$$Return_t < 0, \qquad Return_{t+1} > 0.$$

  A short-lived reversal supports temporary price pressure.

- **Insider trading:** insiders should increase purchases if prices fall below fundamentals.
- **Other-Climate attention:** the response should be stronger after the 2006 Stern Review.
- **Other-Environmental profile:** high EScore and sustainability-oriented investors should reduce selling pressure.

Design–Data

- **Establishments:** 640,203 unique establishments; 75 per firm on average.
- **Stock sample:** 948,595 stock-month observations from 1990 to 2015.
- **Bond sample:** 378,476 bond-month observations from 2004 to 2015.
- **Stock data:** CRSP and Compustat.
- **Bond data:** TRACE Enhanced and Mergent FISD.
- **Environmental and ownership data:** MSCI ESG, 13F, and insider filings.

Result-H1-Future Stock and Bond Returns

Panel A: Stock sample

Dependent variable: *Future Stock Excess Return*

Variable	(1)	(2)	(3)	(4)
Disaster	0.356***		0.146***	
	(0.058)		(0.051)	
Ln(Damage)		0.174***		0.049***
		(0.065)		(0.004)
Ln(BM)			0.446***	0.401***
			(0.128)	(0.143)
Ln(ME)			−1.743***	−1.958***
			(0.037)	(0.040)

Panel B: Bond sample

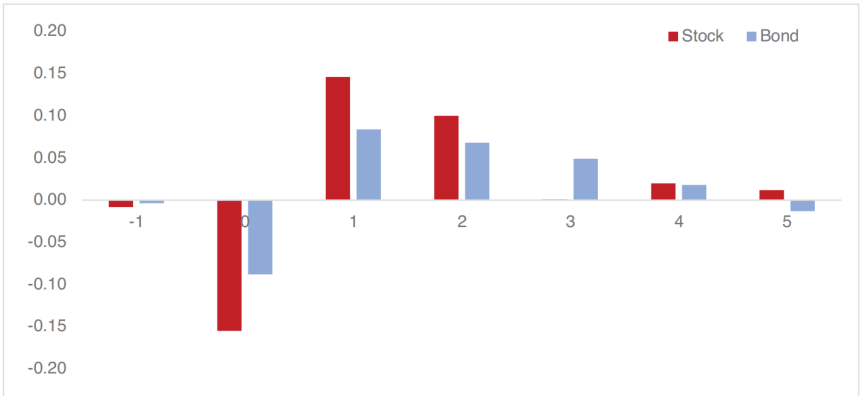
Dependent variable: *Future Bond Excess Return*

Variable	(1)	(2)	(3)	(4)
Disaster	0.150***		0.084***	
	(0.031)		(0.010)	
Ln(Damage)		0.092***		0.052***
		(0.010)		(0.010)
Downside Risk			0.091***	0.090***
			(0.003)	(0.003)
Ln(Maturity)			0.028	0.023
			(0.024)	(0.023)
Ln(Ratin?)			0.383***	0.394***

Result-H2-Contemporaneous and Long-run Returns

Panel A: Stock sample				
Variable	(1) *Stock Excess Return*$_t$	(2) *Stock Excess Return*$_{t+2}$	(3) *Stock Excess Return*$_{t+3}$	(4) *Stock Excess Return*$_{t+4}$
Disaster	−0.155***	0.100**	0.001	0.020
	(0.052)	(0.050)	(0.050)	(0.051)
Control Variables	Yes	Yes	Yes	Yes
Stock fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Number of Obs	954,911	816,321	809,071	801,859
Adjusted R^2	0.023	0.031	0.032	0.022
Panel B: Bond sample				
Variable	(1) *Bond Excess Return*$_t$	(2) *Bond Excess Return*$_{t+2}$	(3) *Bond Excess Return*$_{t+3}$	(4) *Bond Excess Return*$_{t+4}$
Disaster	−0.088***	0.068***	0.049***	0.018
	(0.014)	(0.010)	(0.011)	(0.017)
Control variables	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Number of observations	383,820	373,124	367,622	362,134
Adjusted R^2	0.054	0.070	0.058	0.031

Result-H2-Contemporaneous and Long-run Returns



Result-H3-Insider Trading

Variable	(1) $Pct_Buy_Shares_t$	(2) $Ln(Buy_Shares+1)_t$
Disaster	0.424***	0.043**
	(0.108)	(0.017)
Ln(BM)	−0.595	0.081
	(0.403)	(0.064)
Ln(ME)	0.865***	0.297***
	(0.125)	(0.021)
Reversal	−0.012***	0.004***
	(0.002)	(0.000)
Profitability	−1.716***	−0.259***
	(0.529)	(0.092)
Investment	0.275	−0.066**
	(0.179)	(0.029)
Momentum	0.002**	0.002***
	(0.001)	(0.000)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	954,911	954,911
Adjusted R^2	0.157	0.184

Result-H4-Climate Attention

Panel A: Stock sample

Dependent variable: *Future Stock Excess Return*		
Variable	(1) Jan 1990–Oct 2006	(2) Nov 2006–Dec 2015
Disaster	−0.100	0.426***
	(0.066)	(0.078)
Control variables	Yes	Yes
Stock fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	614,057	334,538
Adjusted R^2	0.026	0.050

Panel B: Bond sample

Dependent variable: *Future Bond Excess Return*		
Variable	(1) Jan 2004–Oct 2006	(2) Nov 2006–Dec 2015
Disaster	0.026	0.031***
	(0.021)	(0.012)
Control variables	Yes	Yes
Bond fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	101,483	276,993
Adjusted R^2	0.093	0.099

Result-H5-Environmental Performance

Panel A: Stock sample

Variable	(1) *Stock Excess Return*$_t$	(2) *Stock Excess Return*$_{t+1}$
Disaster × *EScore*	0.115**	−0.129**
	(0.049)	(0.063)
Disaster	−0.126**	0.283***
	(0.059)	(0.058)
EScore	0.134***	−0.064
	(0.037)	(0.043)
Control variables	Yes	Yes
Stock fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	278,715	276,125
Adjusted R^2	0.037	0.035

Panel B: Bond sample

Variable	(1) *Bond Excess Return*$_t$	(2) *Bond Excess Return*$_{t+1}$
Disaster × *EScore*	0.038***	−0.043***
	(0.014)	(0.006)
Disaster	−0.133***	0.105***
	(0.030)	(0.012)
EScore	0.037**	0.062***
	(0.014)	(0.007)
Control variables	Yes	Yes
Bond fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	292,961	291,850
Adjusted R^2	0.067	0.085

Thanks!

Question & idea?

Replication-H1-Future Stock and Bond Returns

原文：

Panel A: Stock sample

Dependent variable: *Future Stock Excess Return*

Variable	(1)	(2)	(3)	(4)
Disaster	0.356***		0.146***	
	(0.058)		(0.051)	
Ln(Damage)		0.174***		0.049***
		(0.065)		(0.004)
Ln(BM)			0.446***	0.401***
			(0.128)	(0.143)
Ln(ME)			−1.743***	−1.958***
			(0.037)	(0.040)

Panel B: Bond sample

Dependent variable: *Future Bond Excess Return*

Variable	(1)	(2)	(3)	(4)
Disaster	0.150***		0.084***	
	(0.031)		(0.010)	
Ln(Damage)		0.092***		0.052***
		(0.010)		(0.010)
Downside Risk			0.091***	0.093***
			(0.003)	(0.003)
Ln(Maturity)			0.028	0.023
			(0.024)	(0.023)
Ln(Rating)			0.383***	0.394***

复现：

Table 3. Exposure to Natural Disasters and Future Asset Returns

Variable	(1)	(2)	(3)	(4)
Panel A: Stock sample				
	Dependent variable: Future Stock Excess Return			
Disaster	-0.875**		-0.274	
	(0.390)		(0.417)	
Ln(Damage)		-0.117**		-0.025
		(0.053)		(0.056)
Ln(BM)			-0.037	-0.037
			(0.057)	(0.057)
Ln(ME)			-3.792***	-3.792***
			(0.092)	(0.092)
Panel B: Bond sample				
	Dependent variable: Future Bond Excess Return			
Disaster	0.197**		0.257***	
	(0.080)		(0.098)	
Ln(Damage)		0.023**		0.030**
		(0.010)		(0.013)
Illiquidity			35.124	35.124
			(91.386)	(91.386)
Ln(Maturity)			-118.803*	-118.764*
			(67.714)	(67.712)

Replication-H2-Contemporaneous and Long-run Returns

原文：

Panel A: Stock sample

Variable	(1) *Stock Excess Return$_t$*	(2) *Stock Excess Return$_{t+2}$*	(3) *Stock Excess Return$_{t+3}$*	(4) *Stock Excess Return$_{t+4}$*
Disaster	−0.155***	0.100**	0.001	0.020
	(0.052)	(0.050)	(0.050)	(0.051)
Control Variables	Yes	Yes	Yes	Yes
Stock fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Number of Obs	954,911	816,321	809,071	801,859
Adjusted R^2	0.023	0.031	0.032	0.022

Panel B: Bond sample

Variable	(1) *Bond Excess Return$_t$*	(2) *Bond Excess Return$_{t+2}$*	(3) *Bond Excess Return$_{t+3}$*	(4) *Bond Excess Return$_{t+4}$*
Disaster	−0.088***	0.068***	0.049***	0.018
	(0.014)	(0.010)	(0.011)	(0.017)
Control variables	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Number of observations	383,820	373,124	367,622	362,134
Adjusted R^2	0.054	0.070	0.058	0.031

复现：

Table 5. Contemporaneous and Long-Run Market Reactions

Variable	Return t	Return t+2	Return t+3	Return t+4
Panel A: Stock sample				
Disaster	-0.471	1.382***	-0.461	-0.189
	(0.452)	(0.418)	(0.432)	(0.461)
Control variables	Yes	Yes	Yes	Yes
Security fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Number of observations	531,656	517,759	511,521	505,809
Panel B: Bond sample				
Disaster	-0.004***	0.105	0.119	0.020
	(0.001)	(0.100)	(0.088)	(0.105)
Control variables	Yes	Yes	Yes	Yes
Security fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Number of observations	67,412	48,921	46,554	44,266

Replication-H3-Insider Trading

原文:

Variable	(1) $Pct_Buy_Shares_t$	(2) $Ln(Buy_Shares+1)_t$
Disaster	0.424***	0.043**
	(0.108)	(0.017)
Ln(BM)	−0.595	0.081
	(0.403)	(0.064)
Ln(ME)	0.865***	0.297***
	(0.125)	(0.021)
Reversal	−0.012***	0.004***
	(0.002)	(0.000)
Profitability	−1.716***	−0.259***
	(0.529)	(0.092)
Investment	0.275	−0.066**
	(0.179)	(0.029)
Momentum	0.002**	0.002***
	(0.001)	(0.000)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	954,911	954,911
Adjusted R^2	0.157	0.184

复现:

Table 6. Exposure to Natural Disasters and Insider Tradi

Variable	Pct_Buy_Shares t	Ln(Buy_Shares+1) t
Disaster	0.108**	1.247**
	(0.044)	(0.616)
Ln(BM)	-0.061***	-0.562***
	(0.008)	(0.095)
Ln(ME)	-0.073***	-0.728***
	(0.008)	(0.093)
Reversal	-0.331***	-3.917***
	(0.014)	(0.171)
Profitability	0.395***	4.288***
	(0.070)	(0.832)
Investment	0.000***	0.001***
	(0.000)	(0.000)
Momentum	-0.015***	-0.182***
	(0.004)	(0.053)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	44,728	44,768

Replication-H4-Climate Attention

原文：

Panel A: Stock sample

Dependent variable: *Future Stock Excess Return*

Variable	(1) Jan 1990–Oct 2006	(2) Nov 2006–Dec 2015
Disaster	−0.100	0.426***
	(0.066)	(0.078)
Control variables	Yes	Yes
Stock fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	614,057	334,538
Adjusted R^2	0.026	0.050

Panel B: Bond sample

Dependent variable: *Future Bond Excess Return*

Variable	(1) Jan 2004–Oct 2006	(2) Nov 2006–Dec 2015
Disaster	0.026	0.031***
	(0.021)	(0.012)
Control variables	Yes	Yes
Bond fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	101,483	276,993

复现：

Table 7. Impact of Disasters Before and After Januar

Variable	Before Jan. 2016	After Jan. 2016
Panel A: Stock sample		
Disaster	0.816	-1.078**
	(1.144)	(0.490)
Control variables	Yes	Yes
Security fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	165,686	359,189
Panel B: Bond sample		
Disaster	0.027	0.365**
	(0.126)	(0.145)
Control variables	Yes	Yes
Security fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	19,755	32,422

Replication-H5-Environmental Performance

原文：

Panel A: Stock sample

Variable	(1) *Stock Excess Return$_t$*	(2) *Stock Excess Return$_{t+1}$*
Disaster × EScore	0.115**	−0.129**
	(0.049)	(0.063)
Disaster	−0.126**	0.283***
	(0.059)	(0.058)
EScore	0.134***	−0.064
	(0.037)	(0.043)

Panel B: Bond sample

Variable	(1) *Bond Excess Return$_t$*	(2) *Bond Excess Return$_{t+1}$*
Disaster × EScore	0.038***	−0.043***
	(0.014)	(0.006)
Disaster	−0.133***	0.105***
	(0.030)	(0.012)
EScore	0.037**	0.062***
	(0.014)	(0.007)
Control variables	Yes	Yes
Bond fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	292,961	291,850
Adjusted R^2	0.067	0.085

复现：

Table 8. Impact of Disasters and Firm Environmental Performance

Dependent variable	**(1)**	**(2)**
Panel A: Stock sample		
Variable	Stock Excess Return t	Stock Excess Return t+1
Disaster × EScore	2.052***	0.103
	(0.577)	(0.482)
Disaster	0.088	-0.228
	(0.428)	(0.406)
EScore	-0.377***	0.119***
	(0.053)	(0.038)
Control variables	Yes	Yes
Stock fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Number of observations	522,168	515,679
Panel B: Bond sample		
Variable	Bond Excess Return t	Bond Excess Return t+1
Disaster × EScore	-0.002	-0.169
	(0.002)	(0.149)
Disaster	-0.004***	0.222**
	(0.001)	(0.101)
EScore	0.001*	-0.173*
	(0.000)	(0.094)
Control variables	Yes	Yes
Bond fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Month fixed effects	Yes	Yes

Idea

- 中国资本市场应用：基于企业经营地识别洪水、台风、高温等物理气候风险暴露。
  - 识别真正的生产经营地
  - 股债比较：原因在于投资者结构?
- 地方政策改变了企业行为，还是仅仅改变了企业活动发生的位置?
  - 假设某城市出台：环保监管；排放限制；税收政策……
  - 已有很多论文都会做：PolicyCityŒPost.
  - 但是跨地区企业可以把活动转移到其他经营地。